

Separation, Distortion, and Endogenous Timing

A Tullock Contest under Two-sided Asymmetry

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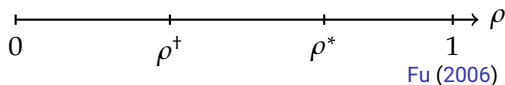
Master's Thesis Defense

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Research Question

- Fu (2006): in a Tullock contest with private information, the informed player prefers to move *second* – revealing type early is costly.
- But this rests on $\rho = 1$: prize values are *perfectly correlated*.
- **Gap**: does the late-mover preference survive at $\rho < 1$, or is it an artifact of full correlation?
- **This paper**: introduce $\rho \in [0, 1]$ – at $\rho = 0$ the signal channel is closed; at $\rho = 1$ Fu's result is recovered as the limiting case.



What This Paper Does

- ***i.i.d.* benchmark** ($\rho = 0$) – isolates timing from information spillover.
- **Partial correlation** ($0 < \rho < 1$) as the main model – $\rho = 1$ is the endpoint, not a separate model.
- Characterizes undistorted vs. Riley distorted separation, and their transition as ρ rises.
- Links subgame equilibria to endogenous timing: a structural asymmetry between I and U .

Related Literature

Strand	Key reference	Main takeaway
Contest success function	Tullock (1980)	Ratio-form CSF; unique equilibrium at $r = 1$
Exogenous timing	Dixit (1987)	Stronger player overcommits first (top-dog)
Endogenous timing	Baik and Shogren (1992)	Weaker player moves first
Timing framework	Hamilton and Slutsky (1990)	SS / UI / IU timing architecture
Asymmetric info (SS)	Hurley and Shogren (1998)	Type uncertainty suppresses U 's effort
Information advantage	Wärneryd (2003)	Informed player can win less

Gap and Contribution

Strand	Key reference	Main takeaway
Signaling in contests	Fu (2006)	$\rho = 1$; late mover avoids revelation cost
Least-cost separation	Riley (1979)	Least-cost separation under binding IC
Timing reversal	Protopappas (2023, 2025)	Early-mover equilibrium under cost asymmetry
Gap	—	<i>i.i.d.</i> $\rightarrow$ fully correlated transition uncharacterized
This paper	—	One ρ spans the spectrum; re-opens the signal channel

Model Setup

- Two players: informed (I) and uninformed (U)
- Types: $V_I, V_U \in \{v_H, v_L\}$, $\Pr(V_k = v_H) = q \in (0, 1)$
- **Benchmark assumption:** $V_I \perp V_U$ (i.i.d., $\rho = 0$)
- Define $\bar{v} \equiv qv_H + (1 - q)v_L = \mathbb{E}[V_k]$

	$V_U = v_H$	$V_U = v_L$
$V_I = v_H$	q^2	$q(1 - q)$
$V_I = v_L$	$q(1 - q)$	$(1 - q)^2$

- Three-stage timing: Stage 0 (choose timing) $\rightarrow$ Nature draws types $\rightarrow$ effort subgame
- Three regimes: SS (simultaneous), UI (U moves first), IU (I moves first)

Equilibrium Payoffs across Regimes

Composite coefficients: $\hat{A} \equiv \bar{v} \mathbb{E}[V_I^{-1/2}]$, $\hat{B} \equiv \bar{v} \mathbb{E}[V_I^{-1}] \geq 1$ (Jensen).

Regime	x_U^*	$\tilde{\pi}_I$	$\tilde{\pi}_U$
SS	$\frac{\hat{A}^2}{(1 + \hat{B})^2}$	$\mathbb{E}\left[\left(\sqrt{V_I} - \frac{\hat{A}}{1 + \hat{B}}\right)^2\right]$	$\frac{\hat{A}^2 \hat{B}}{(1 + \hat{B})^2}$
UI	$\frac{\hat{A}^2}{4}$	$\mathbb{E}\left[\left(\sqrt{V_I} - \frac{\hat{A}}{2}\right)^2\right]$	$\frac{\hat{A}^2}{4}$
IU (sep.)	$\frac{v(2\bar{v} - v)}{4\bar{v}}$	$\frac{\mathbb{E}[V_I^2]}{4\bar{v}}$	$\frac{\mathbb{E}[(2\bar{v} - V_I)^2]}{4\bar{v}}$

→ **First-mover advantage for U:** $\tilde{\pi}_U^{UI} - \tilde{\pi}_U^{SS} = \frac{\hat{A}^2(1 - \hat{B})^2}{4(1 + \hat{B})^2} \geq 0$.

Stage-0 Timing Game

$\tilde{\pi}_U^{UI} \geq \tilde{\pi}_U^{SS}$ always, yet $\tilde{\pi}_I^{SS} > \tilde{\pi}_I^{UI}$ whenever $\hat{B} > 1 - U$ wants to lead, but I refuses to follow: UI is never a pure-strategy equilibrium. The real choice is between IU and SS. Define the informed player's unilateral deviation gaps:

$$D_I \equiv \tilde{\pi}_I^{IU} - \tilde{\pi}_I^{SS}, \quad D_U \equiv \tilde{\pi}_U^{IU} - \tilde{\pi}_U^{SS}.$$

Proposition (Stage-0 Pure-strategy Equilibria, Benchmark)

- UI is never a pure-strategy equilibrium
- IU iff $D_I \geq 0$
- SS iff $D_U \leq 0$
- if $D_I < 0 < D_U$: no pure-strategy equilibrium exists

→ Since $\rho = 0$ closes the signal channel, this split is driven purely by who benefits from committing first – no signaling distortion.

Correlated Type Structure

- Correlation of payoffs: $\rho \equiv \text{corr}(V_I, V_U) \in [0, 1]$
- Marginals fixed at $(q, 1 - q)$ – ρ is the single free parameter of the joint distribution

	$V_U = v_H$	$V_U = v_L$
$V_I = v_H$	$q^2 + \rho q(1 - q)$	$q(1 - q)(1 - \rho)$
$V_I = v_L$	$q(1 - q)(1 - \rho)$	$(1 - q)^2 + \rho q(1 - q)$

ρ	Structure	Key property
$\rho = 0$	<i>i.i.d.</i> benchmark	signal has no spillover
$0 < \rho < 1$	partial correlation	belief enters U 's best response
$\rho = 1$	fully correlated	Fu (2006) endpoint

Posterior Expected Valuation

Lemma (Posterior Expected Valuation)

Given posterior belief $\mu = \Pr(V_I = v_H \mid x_I)$,

$$\tilde{V}_U(\mu) = v_L + [q + \rho(\mu - q)]\Delta V, \quad \frac{d\tilde{V}_U(\mu)}{d\mu} = \rho \Delta V > 0.$$

Posterior μ	Interpretation	$\tilde{V}_U(\mu)$
$\mu = 1$	I is high type	$v_L + [q + \rho(1 - q)]\Delta V$
$\mu = q$	beliefs unchanged	$v_L + q\Delta V = \mathbb{E}[V_U]$
$\mu = 0$	I is low type	$v_L + q(1 - \rho)\Delta V$

→ When $\rho > 0$, I 's type is a signal about U 's own prize – the channel is live.

SS and UI under Partial Correlation

Composite coefficients – replace $\bar{v}$ with $\tilde{V}_U(1), \tilde{V}_U(0)$:

$$\tilde{A} \equiv q \frac{\tilde{V}_U(1)}{\sqrt{v_H}} + (1 - q) \frac{\tilde{V}_U(0)}{\sqrt{v_L}}, \quad \tilde{B} \equiv q \frac{\tilde{V}_U(1)}{v_H} + (1 - q) \frac{\tilde{V}_U(0)}{v_L}.$$

Proposition (First-mover Advantage Robust)

$\tilde{\pi}_U^{UI} \geq \tilde{\pi}_U^{SS}$ for all $\rho \in [0, 1]$.

When Does Separation Hold?

Undistorted separating candidate in the IU subgame, and its IC condition:

$$x_H = \frac{v_H^2}{4\tilde{V}_U(1)}, \quad x_L = \frac{v_L^2}{4\tilde{V}_U(0)}; \quad \frac{\tilde{V}_U(1)}{\tilde{V}_U(0)} \leq R \equiv \frac{v_H^2}{v_L(2v_H - v_L)}.$$

Proposition (Separation Threshold)

Undistorted separation holds iff $\rho \leq \rho^*$, where

$$\rho^* = \frac{(R-1)\tilde{V}_U(q)}{(1-q+Rq)\Delta V}.$$

- At $\rho = 0$: strictly satisfied; at $\rho = 1$: violated
- $\tilde{V}_U(1)/\tilde{V}_U(0)$ is strictly increasing in ρ , so the IC condition reduces to a single threshold ρ^*

Riley Distortion

When $\rho > \rho^*$, define

$$\alpha \equiv 1 - \frac{\tilde{V}_U(0)}{\tilde{V}_U(1)} \in (0, 1), \quad z_H^- \equiv \frac{v_H}{2\sqrt{\tilde{V}_U(0)}}(1 - \sqrt{\alpha}).$$

- High type keeps undistorted effort x_H ; low type distorts *down* to $x_L^R = (z_H^-)^2$

Corollary (Region Structure of the IU Subgame)

- $\rho \leq \rho^*$: undistorted separation – no signaling cost
- $\rho > \rho^*$: Riley distortion – cost borne by the low type alone
- $\rho \geq \rho_P$ may also admit *pooling*: $\mu = q$, the signal is lost – coexists with whichever separating type applies at that ρ

→ As ρ rises, the low type's effort no longer supports separation, so it must distort downward.

Stage-0 Payoff Matrix

Key point: at stage 0, types are unrealized — players compare *ex ante* expected payoffs.

	U: Early	U: Late
I: Early	$(\tilde{\pi}_I^{SS}, \tilde{\pi}_U^{SS})$	$(\tilde{\pi}_I^{IU}, \tilde{\pi}_U^{IU})$
I: Late	$(\tilde{\pi}_I^{UI}, \tilde{\pi}_U^{UI})$	$(\tilde{\pi}_I^{SS}, \tilde{\pi}_U^{SS})$

$$D_I^{UI} \equiv \tilde{\pi}_I^{UI} - \tilde{\pi}_I^{SS}, \quad D_I^{IU} \equiv \tilde{\pi}_I^{IU} - \tilde{\pi}_I^{SS},$$

$$D_U^{UI} \equiv \tilde{\pi}_U^{UI} - \tilde{\pi}_U^{SS}, \quad D_U^{IU} \equiv \tilde{\pi}_U^{IU} - \tilde{\pi}_U^{SS}.$$

Lemma (Stage-0 Pure-strategy Equilibria)

- UI iff $D_I^{UI} \geq 0$ and $D_U^{UI} \geq 0$
- SS iff $D_I^{UI} \leq 0$ and $D_U^{IU} \leq 0$
- IU iff $D_I^{IU} \geq 0$ and $D_U^{IU} \geq 0$

Endogenous Timing under Correlated Type Structure

- U 's side (recap): $D_U^{UI} \geq 0$ for all $\rho \in [0, 1]$ – U always wants to lead against a late-moving I .
- I 's side, new: $D_I^{UI} \leq 0$ throughout the interior, with equality *only* at $\rho = 1 - I$ (almost) never wants to lead via UI .

Theorem (Stage-0 Timing Equilibrium)

- $\rho \leq \rho^+$: IU is the unique equilibrium ($D_I^{IU} \geq 0, D_U^{IU} \geq 0$)
- $\rho^+ < \rho < 1$: IU and UI are both ruled out; SS iff $D_U^{IU}(\rho) \leq 0$; *otherwise no pure-strategy equilibrium exists*
- $\rho = 1$: UI is the unique equilibrium – Fu (2006)'s endpoint

→ The gap beyond ρ^+ is not a clean flip to SS – a band with *no* pure-strategy timing equilibrium can open up (numerical confirmation next).

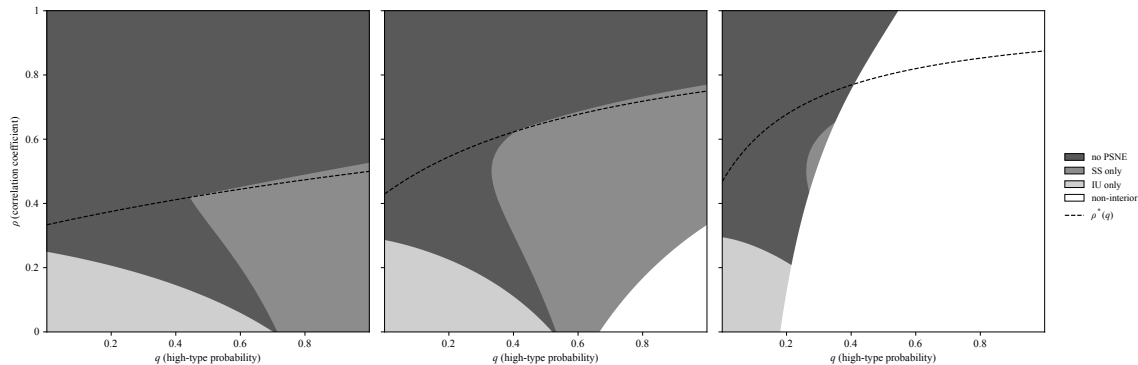
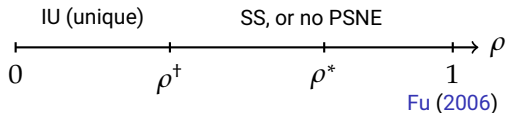


Figure: (q, ρ) Phase Diagram

→ The no-equilibrium band widens, and the IU region's q -range shrinks, as v_H/v_L grows — consistent with ρ^+ falling as the value gap widens.

Summary – The Correlation Spectrum



ρ range	IU subgame (if reached)	Stage-0 outcome
$[0, \rho^+)$	undistorted separation	IU (unique)
$(\rho^+, \rho^*]$	undistorted separation	SS, or no PSNE
$(\rho^*, 1)$	Riley distorted separation	SS, or no PSNE
$\rho = 1$	Fu (2006) endpoint	UI (unique)

→ **Bottom line:** Fu (2006)'s late-mover result *requires* correlation – it is the right endpoint, not a general principle, and the path there is not monotone.

Welfare Framework

Social welfare = allocative value minus effort dissipation: $\widetilde{W} = \widetilde{G} - \widetilde{T}$, where $\widetilde{G} = \mathbb{E}[p_I V_I + p_U V_U]$ and $\widetilde{T} = \mathbb{E}[x_I + x_U]$.

Equilibrium type	Expected dissipation $\widetilde{T}$	Sensitivity to ρ
SS	$\frac{\widetilde{A}}{1 + \widetilde{B}} \mathbb{E}[\sqrt{V_I}]$	ambiguous ($\widetilde{A}$, $\widetilde{B}$ both fall)
UI	$\frac{\widetilde{A}}{2} \mathbb{E}[\sqrt{V_I}]$	strictly decreasing (via $\widetilde{A}$)
IU undistorted sep.	$\frac{\mathbb{E}[V_I]}{2}$	immune to ρ
IU Riley distorted	$\frac{v_H}{2} [1 - (1 - q) \sqrt{\alpha}]$	varies via α

→ Pooling ($\rho \geq \rho_P$) is excluded from the welfare comparison: $D_I^{IU, \text{pool}} \leq 0$ means I never prefers IU under pooling, so it cannot sustain endogenous timing and no separate pooling welfare formula is derived – only IU undistorted separation's dissipation is insulated from ρ .

Three Effects of ρ on Welfare

- **Allocative effect:** $\rho \uparrow \rightarrow \tilde{V}_U(1)$ and $\tilde{V}_U(0)$ diverge $\rightarrow$ win probabilities and prize allocation shift across paths; the welfare effect is not sign-definite a priori
- **Dissipation effect:** regime-dependent — UI dissipation falls with ρ (via $\tilde{A}$), SS is ambiguous; only IU undistorted is *insulated* at $\mathbb{E}[V_I]/2$
- **Threshold effect:** ρ crosses ρ^* $\rightarrow$ Riley distortion begins $\rightarrow$ the low-type path is distorted, but welfare need not jump discontinuously at ρ^*

Key implication

- In IU undistorted separation, $\tilde{W}^{\text{sep}}$ decreases with ρ even though dissipation stays fixed
- Across regimes, welfare rankings remain parameter-dependent: SS and UI react through $\tilde{A}$, $\tilde{B}$, while Riley welfare must be judged from the full formula

Equilibrium vs. Social Optimum

- **SS vs. UI:** welfare ranking is *ambiguous* – sign of $(1 - \tilde{B})(\mathbb{E}[\sqrt{V_I}] - \tilde{A}/2)$; equal when $\tilde{B} = 1$
- **Undistorted sep. vs. Riley:** opposite responses to ρ – $\tilde{W}^{\text{sep}}$ falls while $\tilde{W}^{\text{Riley}}$ rises in ρ – *signal revelation and welfare improvement need not move together*

Root cause

Equilibrium timing is pinned down purely by private payoff incentives; social welfare depends jointly on prize allocation *and* effort dissipation – different criteria. Equilibrium timing is therefore *not* guaranteed to be socially optimal – each regime's ranking must be checked from the full welfare formula.

Conclusion

- 1 $\rho = 0$: unique undistorted separation – I leads at *no revelation cost*.
- 2 $\rho > 0$ reopens the signal–belief–response channel: beliefs enter U 's best response.
- 3 **Separation threshold**: undistorted iff $\rho \leq \rho^*$; beyond, Riley distortion – cost borne by the low type alone.
- 4 **Timing asymmetry**: U 's lead preference is robust; I retreats from leading beyond ρ^+ , opening a band where *no* pure-strategy timing equilibrium survives.
- 5 Fu (2006) is the $\rho = 1$ endpoint – not a general principle.
- 6 **Welfare**: equilibrium timing can depart from the social optimum; responses to ρ are regime-dependent.
- 7 **Policy**: under correlated stakes (lobbying / rent-seeking), staged disclosure helps *iff* IU welfare is higher – else seal it.